

Winter Baektae Meju vs. Summer Deotgyeojang Meju

Process Comparison Table

Microbial Leadership Stage	Winter Baektae Meju Process (Ji Hyun-sook)	Summer Deotgyeojang Meju Process (Jun 3–Jun 28, Actual Data)
Stage 1: Defense & Surface-Film Formation	15% weight loss through surface-drying after shaping. Slowly dries the surface with cold wind until fully dry to the touch, blocking contamination by unwanted microbes at the source.	Approx. 19%–20% weight loss immediately after roasting on a 250°C elvan (baekban-seok) hot stone. • Large meju: 408g→330g (19.1% decrease) • Small meju: 296g→235g (20.6% decrease). ※ The blazing heat of the 250°C elvan stone compresses the surface-drying stage into just 30 minutes, forming a crust like scorched rice (nurungji).
Stage 2: Fungal (Mold) Fermentation	15%–30% weight-loss range. As internal moisture gradually escapes, koji mold (<i>Aspergillus oryzae</i>) penetrates deep inside, producing sweetness through saccharification.	Greenhouse Day 5 through Week 2 (approx. 27%–35% weight-loss range). • Large meju: 297g→265g (total 27.2%→35% decrease) • Small meju: 205g→175g (total 30.7%→40.8% decrease). ※ Daytime warmth inside the sunlit greenhouse rapidly awakens the koji mold, saccharifying the barley starch.
Stage 3: Bacterial Fermentation	30%–40% weight-loss range. After moisture escapes further, <i>Bacillus subtilis</i> and other bacteria take over, developing rich flavor.	Greenhouse Week 3 through just before completion. As the last of the internal moisture concentrates, <i>Bacillus subtilis</i> completes deotgyeojang's characteristic musty, deep, traditional umami flavor.
Stage 4: Final Drying & Completion	42%–47% weight-loss range (end point). After bacterial fermentation, final drying brings the meju to a self-preserving state. ※ For a 2,400g meju, once it reaches 1,370g (42.9% loss) to 1,270g (47.1% loss), it is considered finished and harvested.	Final completion on Day 25 (end point). • Large meju final weight: 218g (total 46.5% loss) • Small meju final weight: 157g (total 46.9% loss). ※ As of June 28, both meju settled into the ~46% final-drying threshold, concluding the process.

[R&D Essay] The Microbial Scale That Transcends the Seasons

The Science of Moisture Control in Winter Baektae Meju and Summer Deotgyeojang Meju

1. Introduction: A Meju's Completion Is Determined Not by the Eye, but by the Scale

In traditional fermentation, the most powerful and honest indicator for controlling microbial activity is not an invisible moisture-percentage figure, but the “weight loss” registered on a scale. Since the solid content does not disappear during the meju-fermenting process, 100% of the weight lost is purely the weight of evaporated moisture.

The traditional “baektae meju,” slowly shaped over two months in winter, and the “summer deotgyeojang meju,” completed in just one month under the harsh June sun where harmful bacteria thrive easily, appear completely different in form, season, and ingredients (soybeans versus barley bran, or deotgyeo).

However, when tracked with a scale through the successive generations of microbes, both meju traced exactly the same moisture-loss curve, arriving at one identical scientific destination.

2. Body: The Four-Stage Weight-Loss Law That Controls the Microbial Relay

① Stage 1 (Defense): Winter Extends Time vs. Summer Compresses Time

The first step of traditional fermentation is “surface-drying” — building a clean defensive barrier so harmful bacteria cannot rot the meju's surface.

Winter baektae meju, after shaping, is slowly dried on the surface using cold winter wind until it loses 15% of its initial weight and becomes dry to the touch.

Summer deotgyeojang meju, however, has no room to proceed through this surface-drying stage slowly, given the extreme risk of spoilage in the hot, humid summer. For this reason, our ancestors buried and roasted the meju in rice husks right after shaping, and more recently, an unconventional method was adopted: roasting the meju on a 250°C elvan (baekban-seok) hot stone. In just 30 minutes, this method forcibly strips away roughly 19%–20% of the initial weight, forming a crust as firm as scorched rice (nurungji) on the surface, while sealing in the core's moisture like a pressure cooker (gelatinizing the starch) and blocking harmful microbes at lightning speed.

② Stage 2 (Fungal Fermentation): The Golden Window for Bright Sweetness (Saccharification)

Once surface defense is complete, the mold — koji — awakens in earnest, extending its mycelium and beginning saccharification. The precise moisture-percentage range in which the mold grows most healthily, undisturbed by harmful bacteria, is honestly revealed on the scale.

For winter baektae meju, in the range where weight decreases from 15% to 30%, koji mold penetrates deep inside, breaking down the soybean's protein and starch into the basic amino acids of umami and sweetness.

This same law is reproduced in summer deotgyeojang meju. Having secured a safety margin through elvan-stone roasting, the meju enters a sunlit, well-ventilated greenhouse, and between Day 5 and Week 2 (a roughly 27%–35% total weight-decrease range), it receives intense daytime warmth and reaches the peak of koji fermentation — the period when the barley bran's starch breaks down vigorously and deotgyeojang's characteristic bright sweetness becomes concentrated.

③ Stage 3 (Bacterial Fermentation): Completing the Umami That Deepens Tradition

Once weight loss reaches the 30%–40% range, internal moisture escapes further, mold activity subsides, and beneficial bacteria — led by *Bacillus subtilis* — take over the relay. Following the path the mold has already opened, these bacteria carry out a second round of protein breakdown, filling both winter baektae meju and summer deotgyeojang meju with the musty yet rich, deep umami flavor unique to traditional Korean fermented sauces (jang).

④ Stage 4 (Final Drying): Entering the Self-Preservation Threshold

Only once fermentation is complete does the meju enter the final drying stage, releasing the last remaining excess moisture so it will not spoil during long-term storage — and only then does the journey truly end.

For a 2,400g winter baektae meju, once it reaches a final weight of 1,370g (42.9% loss) to 1,270g (47.1% loss), it becomes a fully matured meju, ripened all the way to the core.

The summer deotgyeojang meju tracked over 25 days, from June 3 to June 28, likewise showed a remarkable final weight-loss rate: 46.5% for the large meju and 46.9% for the small meju — settling precisely within baektae meju's finishing range of 42%–47%, marking the conclusion of fermentation and drying.

3. Conclusion: The Seasons Differ, but the Law of Microbes Is One

In conclusion, the moisture-loss rate (weight-loss rate) of summer meju and winter meju is nearly identical.

The total amount of weight that must be shed for microbes to safely complete their role, block spoilage bacteria, and reach the self-preserving moisture threshold suitable for long-term storage (air-dried moisture of 12%–14%) converges on 42%–47%, transcending season and ingredient.

Our ancestors simply used winter's cold to stretch the process over two months, reducing moisture at a natural pace, while in summer they avoided heat and contamination by burying the meju in rice husks or, more recently,

compressing the timeline to 25 days through heat control via elvan-stone roasting. Once we understand the microbial metabolic formula hidden behind these differences in outward form, we finally hold the scientific key to bringing traditional fermentation out of the museum and reproducing it perfectly even in a modern living environment.

Ultimately, we should conclude: “Whether summer or winter meju, the final moisture-loss rate (weight-loss rate) at which microbes safely complete fermentation and reach a state ready for long-term storage (self-preservation) is essentially the same, converging on the 42%–47% range.”

Though the outward form — season and ingredients — differs, this deotgyeojang data has shown us that the ecology of microbes and the law of moisture control are, in the end, one and the same.

[R&D Manual] Roasting Deotgyeo Meju Using a Modern Oven

This process reinterprets the traditional method of surface-drying meju on a 250°C elvan (baekban-seok) stone — as used for baekmalsun deotgyeojang — through the oven's enclosed convection heat, achieving a “20% loss of initial weight” and forming a microbial defense barrier in just 20 minutes.

Preparation & Basic Setup

- Oven setting: Preheat to 210°C (convection / hot-air mode recommended)
- Materials needed: Precision scale, oven tray, parchment paper, heat-resistant gloves
- Baseline weight: Measure the meju's weight immediately after shaping (e.g., baseline 400g → target weight 320g)

The Practical 20-Minute “10+10 Alternating” Roasting Process

Step 1: [First Side, 10 min] → Steaming the Interior & Sealed Gelatinization

Action: Line the preheated 210°C oven with parchment paper, place the meju pieces with space between them, and roast for exactly 10 minutes.

Science: The oven's powerful high-temperature hot air instantly begins coating and cooking all six sides of the meju. The moisture inside is trapped and cannot escape, creating a “pressure-cooker effect” that steams the interior moist, and preparing the barley starch to gently break down (gelatinize) into a form microbes can easily consume.

Step 2: [Flip & 10 min] → Surface Roasting & Settling at the 20% Golden Weight-Loss Line

Action: After 10 minutes, open the oven, quickly flip the meju, and roast for another 10 minutes.

Science: The side that was slightly damp from contact with the oven tray now faces the hot air directly, and the entire surface transforms into a firm, savory “clean defense barrier” like scorched rice. As steam honestly escapes through the oven's fine vents, the weight drops noticeably.

Oven Harvest & Scale Verification (Most Important Step)

Immediately after finishing: Once the full 20 minutes of roasting is complete, remove the meju immediately and place it on a scale.

- 20% loss achieved (success): If a 400g meju has dropped to roughly 320g, the oven's ventilation efficiency and the meju's thickness were a perfect match. Let it cool as-is and send it to the greenhouse (fermentation room).
- If moisture loss is insufficient (oven seals too tightly): If the weight has only dropped to around 340g (15% loss), leave the oven door cracked open about 1cm (to force steam out) and run it for another 3–5 minutes; it will settle honestly at the 20% line.

[R&D Manual] Roasting Deotgyeo Meju Using a Frying Pan

Since an ordinary frying pan has completely different heat-transfer properties from an elvan stone, cooking the meju perfectly through to the center without burning requires precise heat-control techniques specific to frying pans.

Limitations of Ordinary Frying-Pan Roasting & How to Overcome Them

Unlike elvan stone, an ordinary steel or coated frying pan does not emit far-infrared radiant heat; heat transfers only linearly to the surface. If heat is not controlled properly, the meju easily ends up “burnt black on the outside, raw and damp on the inside.”

The Golden Manual for Frying-Pan-Roasted Deotgyeojang Meju

Step 1: Push Heat to the Core on Low Heat (15 min) — Lid closed, lowest heat

Place the dough on a dry frying pan with no oil, and start on the lowest possible heat with the lid firmly closed. Keeping the lid closed allows moisture inside the dough to evaporate and gently steam the meju's interior. Flip it back and forth gently to cook the interior starch first.

Step 2: Induce the Maillard Reaction on Medium-Low Heat (10 min) — Lid open, medium-low heat

Once the interior is cooked enough and the meju has firmed up, open the lid and raise the heat slightly to medium-low. This is the stage of driving off surface moisture in earnest and inducing the “Maillard reaction,” which produces a savory, scorched-rice aroma. Watch closely and flip frequently to prevent burning.

Step 3: Dry the Surface & Reduce Moisture Using Residual Heat (5 min) — Turn off heat, cool in the pan

Once the surface turns a golden, pleasant chestnut-brown color, turn off the heat. Leave the meju as-is, using the pan's remaining residual heat, until it reaches 20% weight loss (final moisture around 35%) and the steam has fully dissipated, then remove it.

Ji Hyun-sook's Practical Tips

- Adjusting meju thickness: When roasting in a frying pan, shaping it as thick as a master craftsman would means the inside will never cook through. For home use, shape it flat and thin like a barley flatcake (about 1.5–2cm thick) so the pan's conductive heat can reach all the way to the center.
- Using a thick pan: A thin pan, like a cheap aluminum pot, will burn the moment the meju is placed on it. Using the thickest, heaviest pan available at home — such as a cast-iron pan or a fully-clad 5-ply stainless steel pan — works much better for producing gentle, elvan-stone-like heat.